

STAGE 3 FINAL PRESENTATION

GitHub Portfolio & Research Proposal

Research topic: Inflation Targeting and Financial Stability

Selected Emerging and Developed Economies, 1995–2022

5-minute presentation

GitHub + Proposal

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MBAI 5310G – AI Programming · Instructor: Zahra Atf · June 2026

Goal: show GitHub evidence first, then proposal topic, gap, method and contribution.



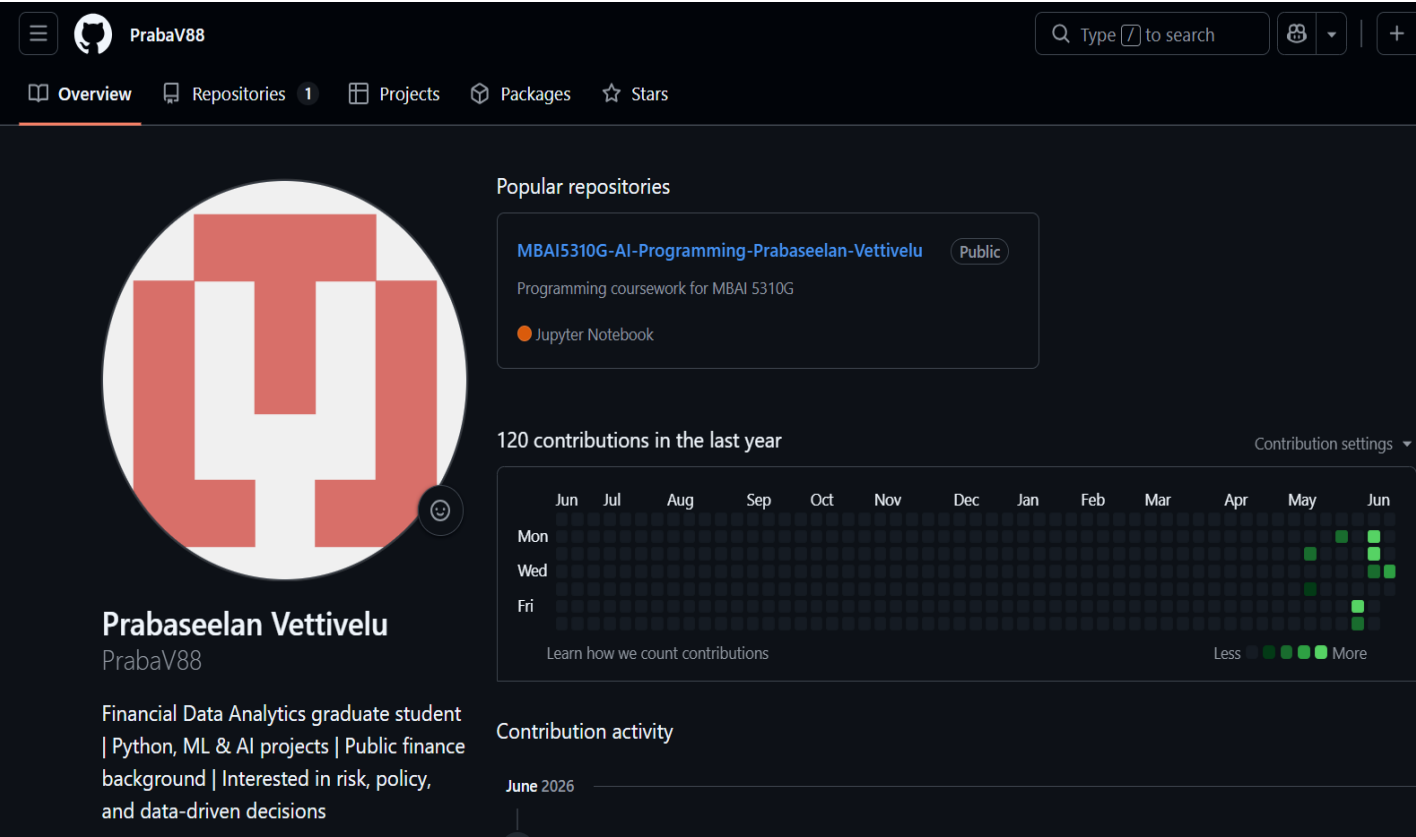
PART A

GitHub Portfolio

Repository organization, coursework evidence, and reproducibility

01

1. Repository overview



WHAT THE REPOSITORY DEMONSTRATES

- **Public course repository** for MBAI 5310G – AI Programming
- Assignment folders, README documentation, reports and outputs
- Shows reproducibility, organization and responsible academic data use

119

commits

5+

coursework folders

1

research folder

Live demo path: Main repo → Research_Proposal → Stage 2 proposal → conceptual framework

2. Repository structure and files

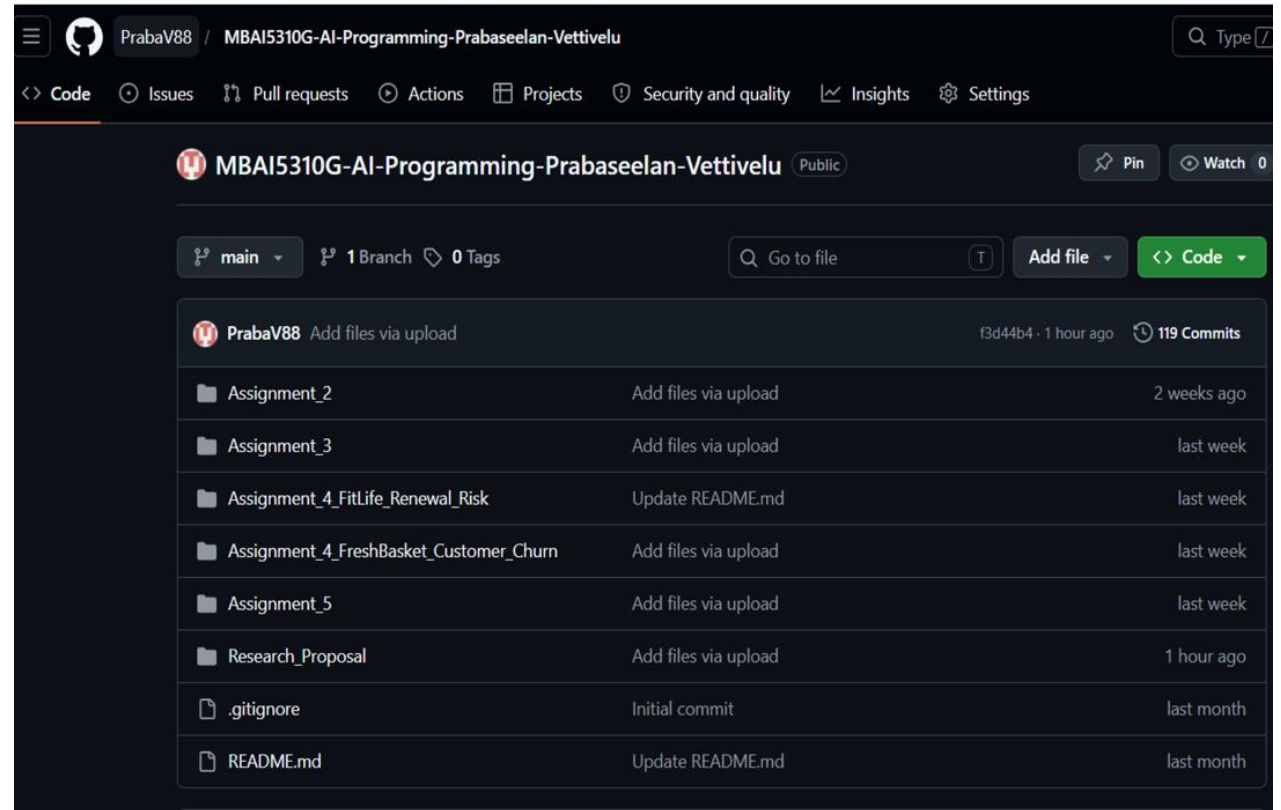
Current organization

- Assignment_2 — end-to-end ML pipeline
- Assignment_3 — Logistic Regression vs SVM
- Assignment_4_FitLife — Decision Tree risk prediction
- Assignment_4_FreshBasket — churn prediction
- Assignment_5 — K-means clustering
- Research_Proposal — Stage 1, Stage 2, framework and slides

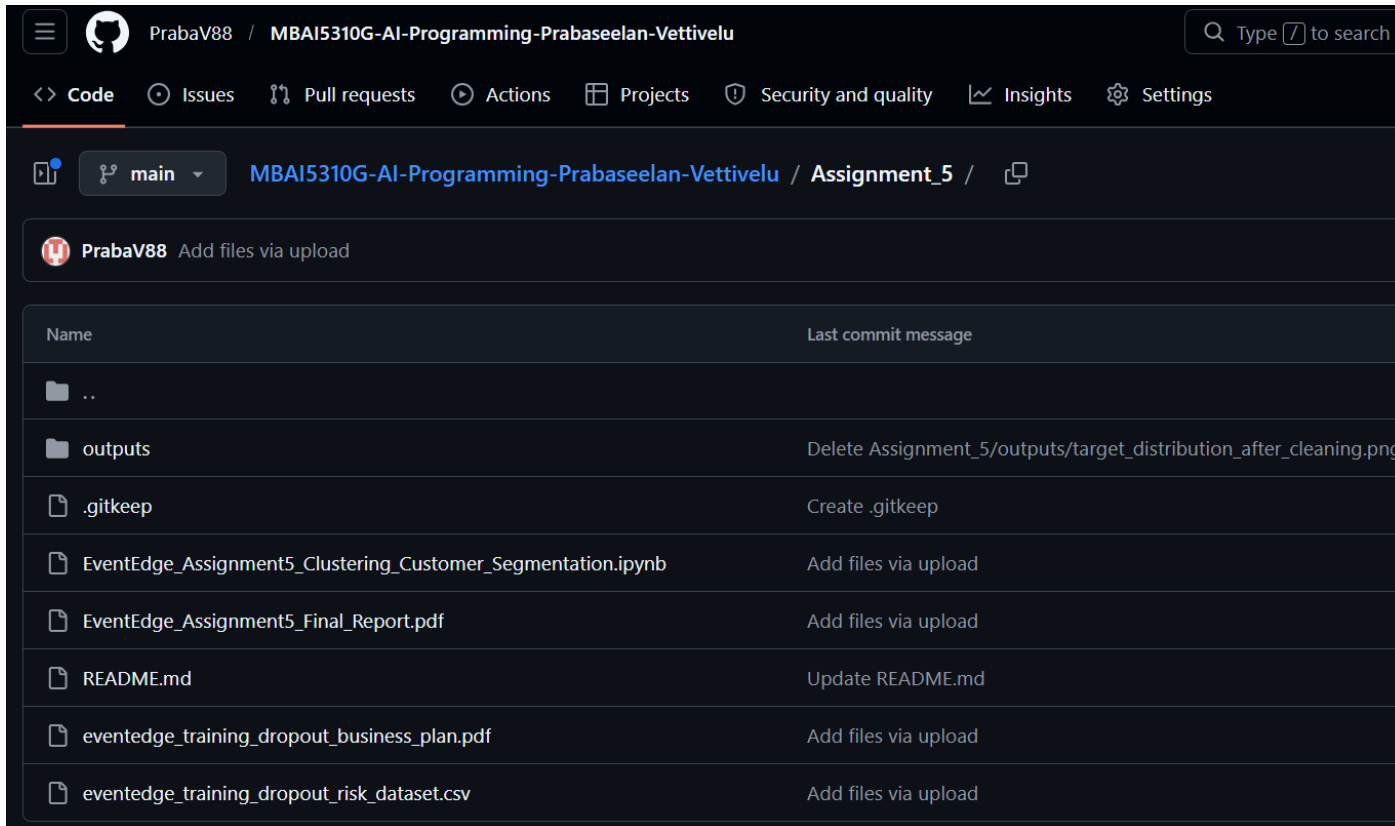
Each assignment folder includes:

Notebook · Dataset · README · Report · Outputs

This structure makes the repository easy to navigate and reproduce.



3. Example assignment evidence: Assignment 5



WHY I WOULD SHOW ASSIGNMENT 5 LIVE

- Latest assignment using **unsupervised learning**
- Notebook, dataset, report, README and outputs are visible
- **K-means clustering and PCA** align with the final proposal methodology
- Business interpretation and responsible AI reflection are documented

PRESENTATION LINE

“This folder shows my AI programming workflow from data preparation to clustering outputs and business interpretation.”

GitHub part takeaway: the repo supports documentation, reproducibility and review.

PART B

02

Research Proposal

Topic, gap, methodology, contribution, timeline, and key references

Inflation Targeting and Financial Stability in Selected Emerging and Developed Economies, 1995–2022: A Data Science-Based Study Using Four Stability Indicators

4. Background and problem statement

Why this topic matters

- Inflation targeting can improve **credibility, transparency and inflation expectations**
- **Price stability does not automatically guarantee financial stability**
- Financial stability includes banking, credit, exchange-rate and inflation pressures

Main problem

The relationship between inflation targeting and financial stability remains unclear.

WHY ONE INDICATOR IS NOT ENOUGH

- Some studies find banking stability benefits
- Others find credit booms, exchange-rate pressure or fragility
- The proposal uses four financial-stability indicators to classify risk

Banking stability

Credit growth

Exchange-rate volatility

Inflation volatility

Proposal response: build a country-year risk label and use AI/data science to classify and interpret patterns.

5. Research gap

1

Mixed evidence

Prior studies do not agree whether inflation targeting is stabilizing or risky.

2

Measurement gap

Many studies focus on one channel instead of multiple stability indicators.

3

Country comparison gap

Emerging and developed economies differ in institutions and shock exposure.

4

Methodological gap

Traditional regressions may miss nonlinear and classification-based risk patterns.

5

Data-structure issue

Some early developed targeters have limited within-sample IT variation.

This study addresses the gap by combining four indicators with classification, clustering, evaluation and interpretation.

Key message: the proposal is finance-first; AI/data science is used as the analytical method.

6. Aim and research questions

AIM

To examine the relationship between inflation-targeting monetary policy and financial-stability risk in selected emerging and developed economies from 1995 to 2022, using a data science-based methodology.

CORE FINANCE QUESTIONS

- How is **inflation targeting** related to **financial-stability risk**?
- How does the **relationship differ across emerging and developed economies**?
- Which **stability indicators** show clearer associations?

AI/DATA SCIENCE QUESTIONS

- Can classification identify high-risk country-years?
- Can clustering reveal financial-stability profiles?
- Which variables explain the risk patterns?

The AI part is not a new topic — it is the method used to study inflation targeting and financial stability.

7. Brief literature review

A

Inflation targeting foundation

Bernanke & Mishkin; Mishkin & Schmidt-Hebbel — credibility, transparency and inflation performance.

B

Supportive stability evidence

Fazio et al.; Mishra & Dubey — banking stability and spillover effects.

C

Cautionary evidence

Fouejieu; Frappa & Mésonnier; Akram & Eitrheim — fragility, credit, housing and asset-price risks.

D

Context and policy conditions

Fazio et al.; Kim & Mehrotra; Mugrabi et al.; Woodford — institutions, macroprudential policy and flexible IT.

Literature conclusion

Inflation targeting is neither uniformly stabilizing nor uniformly destabilizing. Its relationship with financial stability depends on indicator, country type, institutional quality and macroprudential policy.

8. Conceptual framework

HOW THE STUDY IS FRAMED

- Inputs: inflation targeting status, country type, institutional quality and policy indicators
- Outcome: financial-stability risk based on four indicators
- Two analysis paths: supervised classification and unsupervised clustering
- Interpretation remains **predictive and associative, not causal**

FOUR STABILITY INDICATORS

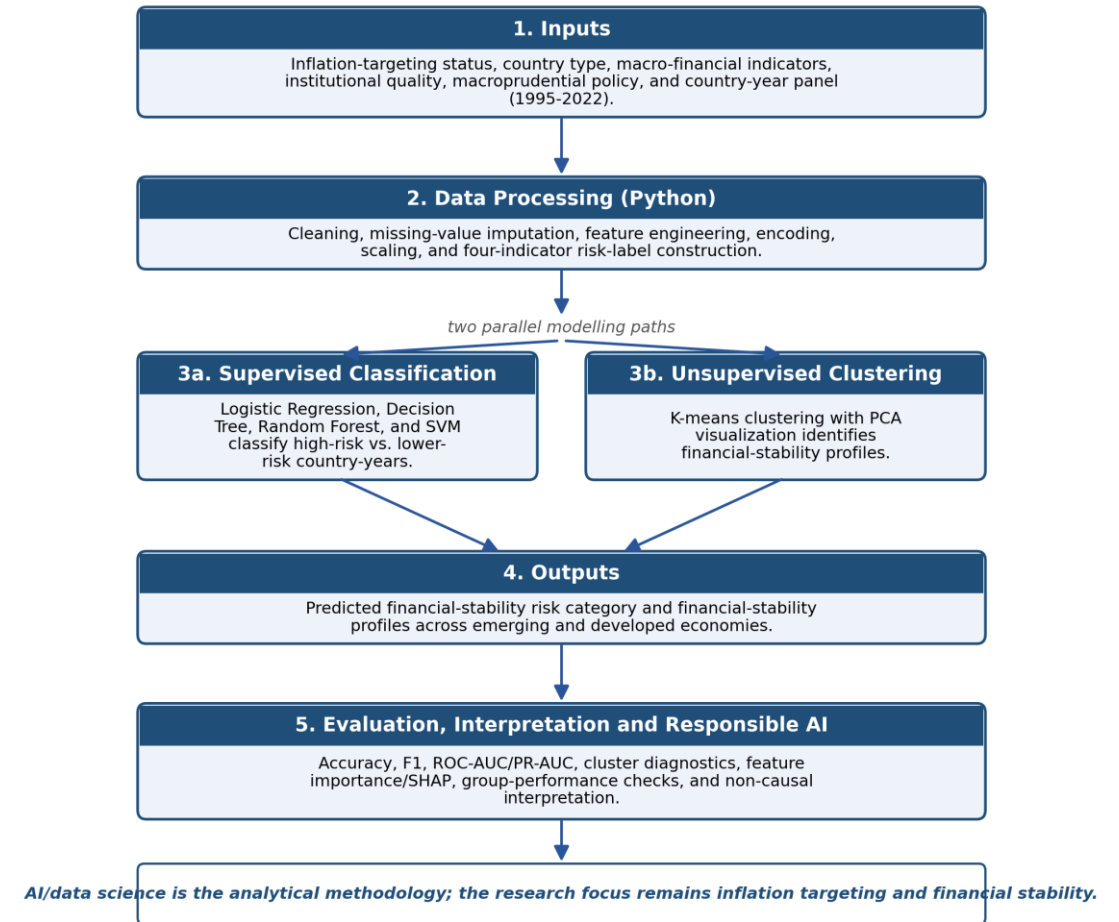
Banking stability

Credit growth

Exchange-rate volatility

Inflation volatility

AI/data science is used as the analytical methodology; the research focus remains inflation targeting and financial stability.



9. Methodology: how the study can be completed

Data	Risk label	Models	Validation	Explain
Up to 20 economies 1995–2022 country-years	Four indicators 25–30% high-risk cut-off	Logistic Regression Decision Tree · RF · SVM	Accuracy · F1 · ROC-AUC PR-AUC · grouped checks	Feature importance SHAP/LIME if feasible

TWO MODELLING PATHS

- Supervised classification: predict high vs lower risk country-years
- Unsupervised clustering: identify financial-stability profiles using K-means + PCA
- Leakage control: same-year stability indicators used to create the label will not be used as predictors

RESPONSIBLE INTERPRETATION

- Secondary country-level data only
- Results are exploratory, predictive and associative
- Small-sample caution and group performance checks will be reported

Tools: Python · pandas · scikit-learn · Jupyter/Colab · GitHub

10. Expected contribution, timeline and key references

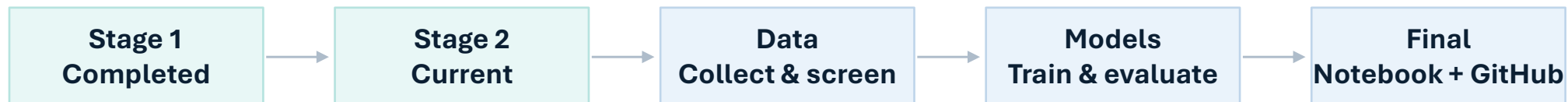
EXPECTED CONTRIBUTION

- Academic: connects inflation-targeting literature with risk classification
- Practical: supports macro-financial risk screening
- Methodological: reproducible Python workflow with explainability and responsible AI

KEY REFERENCES

Bernanke & Mishkin (1997)
Mishkin & Schmidt-Hebbel (2007)
Fazio et al. (2015, 2018)
Fouejieu (2017)
Frappa & Mésonnier (2010)
Woodford (2012); Hammond (2012)
Mugrabi et al. (2026); Ngepah et al. (2026)

PROPOSED COMPLETION PATH



Thank you